EXPERIMENT:6

PAL THAKOR(23BME002)

1. Aim

To compare the Jacobi and line-TDMA/SOR numerical solutions of the steady two-dimensional Laplace equation for heat conduction in a rectangular plate, and to assess their convergence behaviour and accuracy by comparison with the analytical solution.

2. Problem Description

Steady heat conduction in a homogeneous plate with no internal heat generation is represented by the two-dimensional Laplace equation. The plate occupies a rectangular domain of length L and height H. The left boundary is maintained at a sinusoidal temperature distribution, while the right, top and bottom boundaries are maintained at zero temperature.

3. Governing Equation

$$\partial^2 T/\partial x^2 + \partial^2 T/\partial y^2 = 0$$

where T is temperature and x and y are the spatial coordinates.

4. Boundary Conditions

Boundary	**Mathematical condition**
Left boundary (x = 0)	T(0, y) = T_left sin(πy/H)
Right boundary (x = L)	T(L, y) = 0
Bottom boundary (y = 0)	T(x, 0) = 0
Top boundary (y = H)	T(x, H) = 0

5. Analytical Solution

The analytical reference solution used by the supplied solver is:

$$T(x,y) = T_{left} \sinh[\pi(L - x)/H] \sin(\pi y/H) / \sinh(\pi L/H)$$

The numerical temperature fields are compared with this analytical field using the relative L_2 error.

6. Numerical Methodology

6.1 Finite-Difference Discretization

The plate is divided into a uniform Cartesian grid. The Laplace equation at an interior grid point is approximated using central differences:

$$(T_{i+1,j} - 2T_{i,j} + T_{i-1,j})/\Delta x^2 + (T_{i,j+1} - 2T_{i,j} + T_{i,j-1})/\Delta y^2 = 0$$

Rearranging gives the weighted five-point update used by the iterative solver. The grid-spacing ratio is $\beta = (\Delta x/\Delta y)^2$.

6.2 Jacobi Method

In the Jacobi method, all interior temperatures for the next iteration are calculated using only values from the previous iteration. The boundary values are then reapplied. This method is straightforward but generally converges relatively slowly because information is propagated through the grid over many iterations.

6.3 Line-TDMA/SOR Method

The line method solves a tridiagonal system along each horizontal interior grid line using the Thomas algorithm (TDMA). The new line solution is combined with the previous value using the relaxation relation:

$$T_{new} = (1 - \omega)\, T_{old} + \omega\, T_{TDMA}$$

Here, ω is the relaxation factor. In this run, $\omega = 1.55$. A relaxation factor greater than 1 applies over-relaxation and can accelerate convergence when appropriately selected.

7. Computational Parameters

Parameter	Value used
Run mode	lab
Number of grid points in x, nx	81
Number of grid points in y, ny	61
Plate length, L	1.0
Plate height, H	1.0
Left-boundary temperature scale, T_left	100
Maximum iterations	10,000
Convergence tolerance	1×10^{-8}
SOR relaxation factor, ω	1.55
Residual print interval	1,000 iterations
Grid spacing, $\Delta x = L/(nx-1)$	0.0125
Grid spacing, $\Delta y = H/(ny-1)$	0.016667 (approx.)

The supplied lab configuration and solver parameters are documented in the experiment notebook.

8. Results and Discussion

8.1 Quantitative Comparison

Method	Iterations	Final residual ($L\infty$ update)	Relative L_2 analytical error
Jacobi	10,000	1.621069×10^{-6}	7.893996×10^{-5}
Line-TDMA/SOR	938	9.905264×10^{-9}	1.055616×10^{-4}

The line-TDMA/SOR method reached the specified stopping tolerance of 1×10^{-8} in 938 iterations. Jacobi reached the maximum allowed 10,000 iterations, but its final update residual remained 1.621069×10^{-6}, which is above the specified tolerance. Therefore, Jacobi did not satisfy the stopping criterion within the iteration limit.

The line-TDMA/SOR method required about 10.7 times fewer iterations than the 10,000-iteration Jacobi run. However, the Jacobi result has a slightly smaller reported relative L_2 analytical error (7.893996×10^{-5}) than the line-TDMA/SOR result (1.055616×10^{-4}). Thus, the method with the smaller stopping residual is not automatically the one with the smaller analytical error in this particular run.

8.2 Horizontal Centreline Temperature Profile

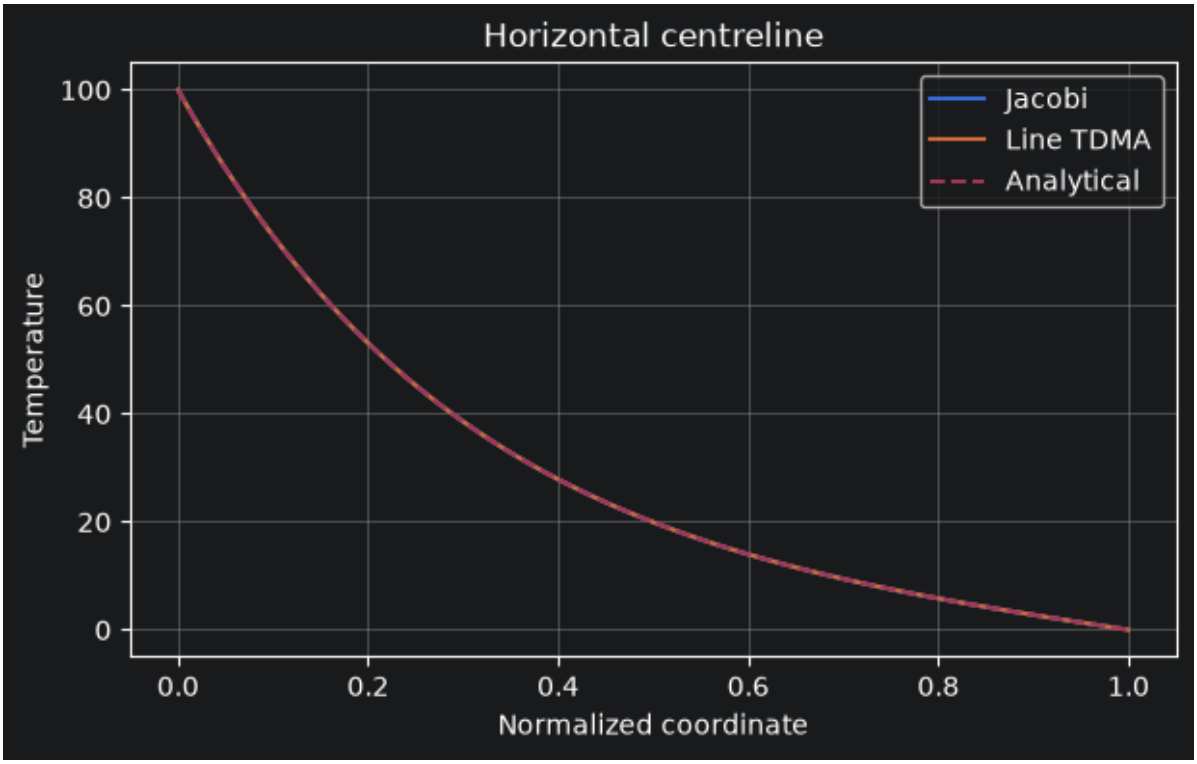


Figure 1. Horizontal centreline temperature distribution: Jacobi, line-TDMA and analytical solution.

The three curves almost overlap throughout the normalized horizontal coordinate. Temperature decreases from approximately 100 at the heated left boundary to zero at the right boundary. The close overlap indicates that both numerical methods reproduce the analytical centreline profile closely at the plotted scale.

8.3 Convergence History

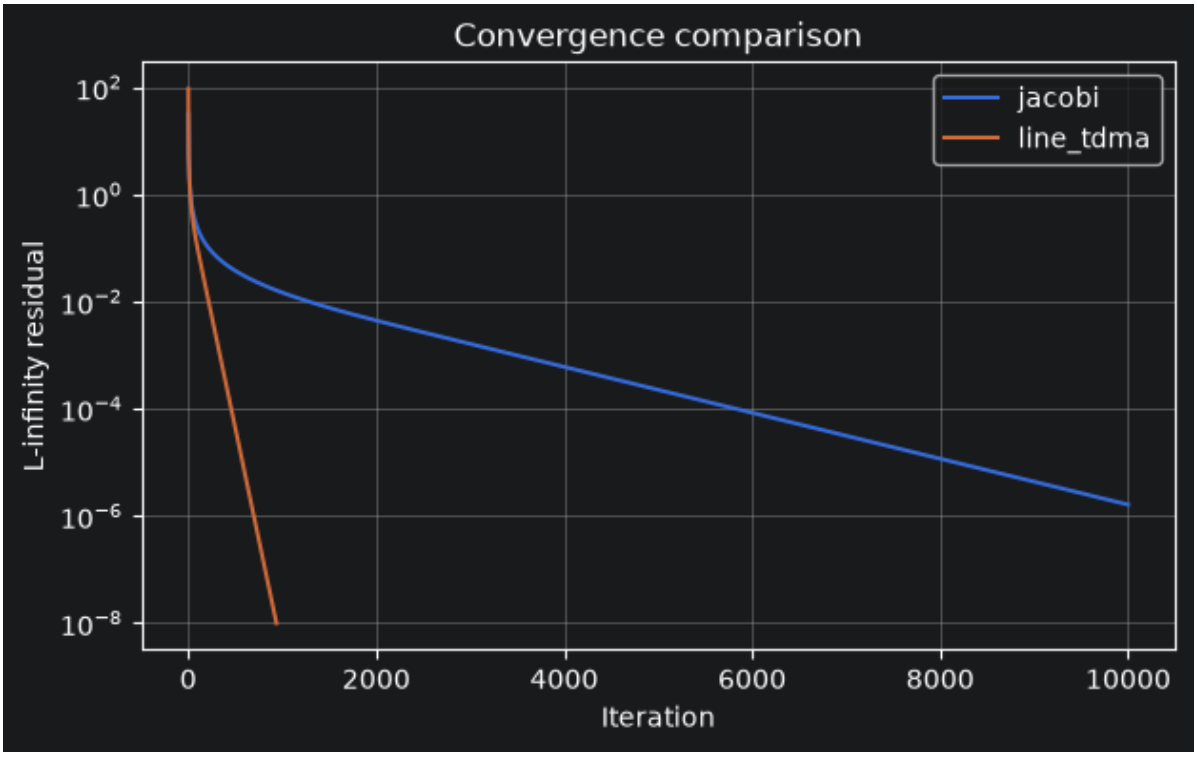


Figure 2. L-infinity iteration-update residual versus iteration count (logarithmic vertical scale).

The residual decreases for both methods. The line-TDMA/SOR curve falls much more steeply and reaches approximately 10^{-8} in fewer than 1,000 iterations. The Jacobi residual decreases more gradually and is still approximately 1.62×10^{-6} at 10,000 iterations. This confirms the faster convergence observed for line-TDMA/SOR in the tested case.

8.4 Two-Dimensional Temperature Field

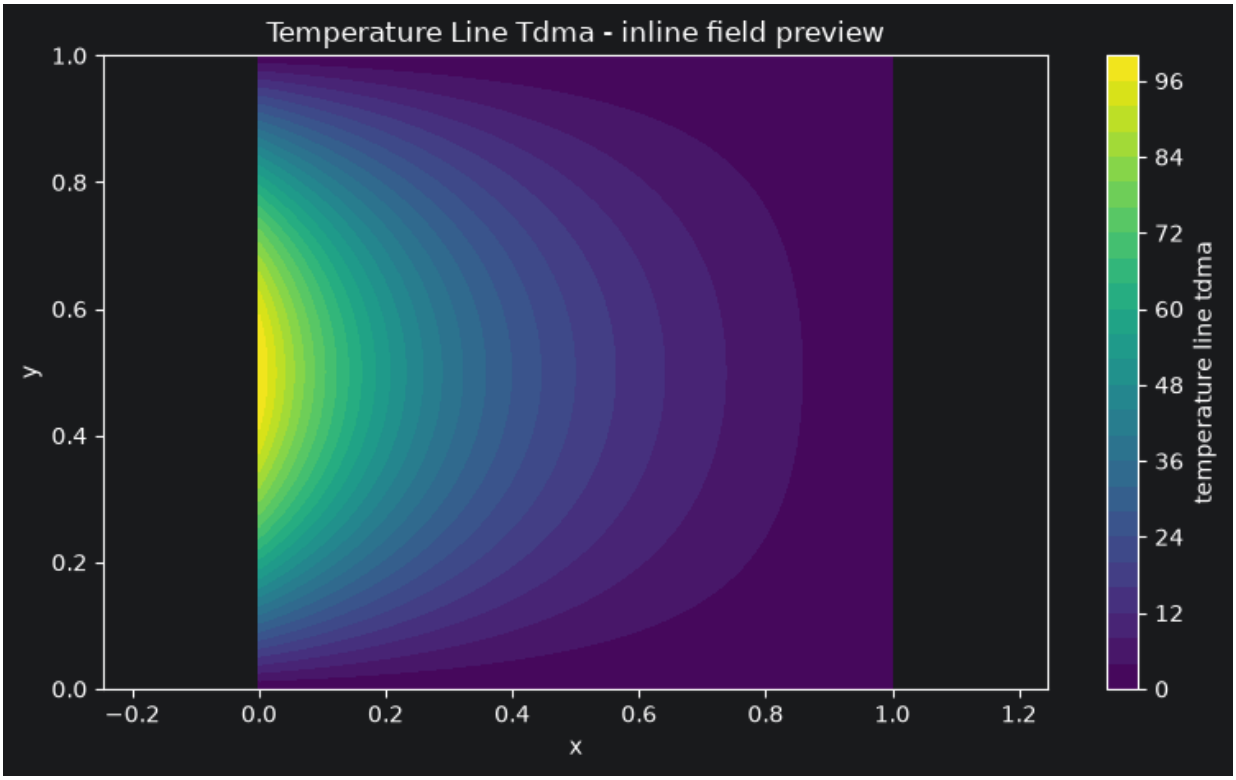


Figure 3. Two-dimensional temperature contour obtained using line-TDMA/SOR.

The highest temperature occurs near the middle of the heated left boundary. The temperature decreases from the left side towards the right side and also approaches zero at the top and bottom boundaries. The smooth contour pattern is consistent with heat spreading through the plate from the prescribed hot boundary towards the zero-temperature boundaries.

9. Effect of Relaxation Factor

The relaxation factor used in the reported line-TDMA/SOR run was $\omega = 1.55$. The solver applies this factor after solving each tridiagonal line system. The present output contains results for this one relaxation-factor value only. Therefore, a quantitative conclusion about how changing ω affects iteration count or residual cannot be made from the supplied run. To complete this part of the compulsory assignment, repeat the same grid and tolerance with at least two additional values of ω (for example, $\omega = 1.0$ and $\omega = 1.3$), then compare the iteration counts and final residuals. These additional values are suggested test cases, not measured results.

10. Grid-Dependence Requirement

The supplied result is for one grid only: 81 × 61 points. The compulsory assignment asks for a comparison on at least two grids. A second-grid result was not included in the supplied output, so no second-grid iteration count, residual or analytical error is reported here. For completion, run another grid (for example, 41 × 31 points or a finer approved grid) using the same physical dimensions, boundary conditions, tolerance and relaxation factor. Record its measured results in a second comparison table.

Grid	**Jacobi iterations / residual / error**	**Line-TDMA/SOR iterations / residual / error**
81 × 61	10,000 / 1.621069×10^{-6} / 7.893996×10^{-5}	938 / 9.905264×10^{-9} / 1.055616×10^{-4}
Second grid	Run required — data not supplied	Run required — data not supplied

11. Observed Numerical Behaviour and Physical Interpretation

Numerical observations: (1) the line-TDMA/SOR iteration converged to the specified update tolerance much faster than Jacobi; (2) the Jacobi run reached its maximum iteration count without meeting the tolerance; (3) the centreline profiles from both methods closely follow the analytical curve; and (4) the reported relative L_2 errors are both small, with the Jacobi value slightly lower for this run.

Physical interpretation: the imposed temperature is highest along the central part of the left edge and zero along the other three edges. Heat therefore conducts away from the hot boundary into the plate, while the top, bottom and right boundaries remain at zero temperature. The contour plot shows this spatial decrease in temperature.

The residual plotted by the solver is the maximum absolute change in the temperature field between successive iterations, not a separately evaluated dimensional heat-flux error. The analytical error is a different measure: it compares the numerical field with the analytical reference solution.

12. Limitations of the Teaching Model

- The model is steady-state; it does not calculate transient temperature changes with time.
- The plate is represented by a two-dimensional domain and does not include through-thickness temperature variation.
- The governing model assumes constant, homogeneous material properties and no internal heat generation.
- The supplied case uses prescribed boundary temperatures; convection, radiation and contact thermal resistance are not modelled.
- Only one grid and one relaxation factor are represented in the supplied numerical output, so the required grid-dependence and relaxation-factor studies remain incomplete.
- The reported convergence residual is based on the change between iterations. It should not be interpreted directly as the physical temperature error or heat-transfer rate.

13. Conclusion

The steady two-dimensional plate conduction problem was solved numerically using the Jacobi and line-TDMA/SOR methods on an 81 × 61 grid. Both numerical centreline profiles closely matched the analytical solution. The line-TDMA/SOR method converged to the prescribed tolerance in 938 iterations, whereas Jacobi did not reach the tolerance within 10,000 iterations. This demonstrates the faster convergence of the line-implicit method for the tested case. The relative L_2 analytical errors were 7.893996×10^{-5} for Jacobi and 1.055616×10^{-4} for line-TDMA/SOR. A second grid and additional relaxation-factor runs are still needed to fully satisfy the compulsory assignment.